

Digital Resonance Model — Math-Clean Specification

1) Constants

Let $\backslash \mathbf{n}$

$$\backslash \mathbf{n} \mathcal{A} = \{\alpha_0, \alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5, \alpha_6, \alpha_7\} \backslash \mathbf{n}$$

be positive real parameters with the following exact definitions where available:

- $\alpha_0 = 1$.
- $\alpha_1 = T$, the unique real root > 1 of $x^3 = x^2 + x + 1$. Equivalently, T is the Tribonacci constant, i.e., the real root of $x^3 - x^2 - x - 1 = 0$.
- $\alpha_2 = \varphi = \frac{1+\sqrt{5}}{2}$.
- $\alpha_3 = \frac{1}{2}$.
- $\alpha_4 = \frac{1}{2\pi}$.
- $\alpha_5 = 2\pi$.
- $\alpha_6 \in \mathbb{R}_{>0}$ (user-specified real constant).
- $\alpha_7 = \text{Im}(\rho_1)/1000$, where ρ_1 is the first nontrivial zero of $\zeta(s)$ on the critical line $\Re s = \frac{1}{2}$.

Immediate identity: $\alpha_4 \alpha_5 = 1$.

2) Resonance map

For a bitstring $b = (b_0, \dots, b_7) \in \{0, 1\}^8$, define the resonance value

$$R(b) = \prod_{i=0}^7 \alpha_i^{b_i} \in \mathbb{R}_{>0}.$$

The *resonance set* is $\mathcal{R} = \{R(b) : b \in \{0, 1\}^8\}$.

3) Canonical form via the $(4, 5)$ -pair

Introduce the integer

$$e(b) := b_5 - b_4 \in \{-1, 0, 1\}.$$

Using $\alpha_4 \alpha_5 = 1$, every resonance value can be written in the *canonical form*

$$R(b) = \alpha_1^{b_1} \alpha_2^{b_2} \alpha_3^{b_3} \alpha_6^{b_6} \alpha_7^{b_7} \alpha_4^{-e(b)} \quad \text{with } b_0 \in \{0, 1\}, e(b) \in \{-1, 0, 1\}.$$

Thus the pair (b_4, b_5) contributes only the three possibilities $e = -1, 0, 1$ rather than four distinct values.

4) Cardinality of the image $\mathcal{R}$

Let the *free* bits be $\{1, 2, 3, 6, 7\}$ (five positions), the neutral bit 0, and the $(4, 5)$ -pair compressed to $e \in \{-1, 0, 1\}$. Then $|\mathcal{R}| = 2^5 \cdot 3 = 96$. *Reason:* there are 2^5 independent choices for b_1, b_2, b_3, b_6, b_7 , three distinct outcomes for e , and b_0 does not change the value since $\alpha_0 = 1$.

5) Degeneracy structure

Define an equivalence relation on $\{0, 1\}^8$ by $b \sim b' \iff R(b) = R(b')$. Two independent symmetries act:

1. Toggle b_0 (multiplication by $\alpha_0 = 1$).
2. Toggle b_4 and b_5 simultaneously (multiplication by $\alpha_4 \alpha_5 = 1$).

Hence, each class has size 2 or 4 : - If $e = 0$ (i.e., $(b_4, b_5) \in \{(0, 0), (1, 1)\}$), the class has size 4 (toggle b_0 ; toggle b_4, b_5). - If $e = \pm 1$, the class has size 2 (only the b_0 toggle preserves e).

Counting classes: for each of the $2^5 = 32$ assignments of the free bits, there is one $e = 0$ value and two $e = \pm 1$ values. Therefore

$$\#\{\text{degeneracy } 4\} = 32, \quad \#\{\text{degeneracy } 2\} = 64, \quad 32 \cdot 4 + 64 \cdot 2 = 256.$$

6) Closed-form aggregate identities

All sums and products over $\{0, 1\}^8$ factor by independence of bits.

6.1 Sum over all 256 states

$$S_{256} := \sum_{b \in \{0, 1\}^8} R(b) = \prod_{i=0}^7 (1 + \alpha_i).$$

Using $\alpha_0 = 1$ and $\alpha_4 \alpha_5 = 1$, one convenient exact form is

$$S_{256} = 2(1 + \alpha_1)(1 + \alpha_2)(1 + \alpha_3)(1 + \alpha_6)(1 + \alpha_7) \left(2 + 2\pi + \frac{1}{2\pi}\right).$$

6.2 Product over all 256 states

$$P_{256} := \prod_{b \in \{0, 1\}^8} R(b) = \prod_{i=0}^7 \alpha_i^{2^7} = \left(\prod_{i=0}^7 \alpha_i\right)^{128}.$$

Since $\alpha_4 \alpha_5 = 1$ and $\alpha_0 = 1$, this reduces to

$$P_{256} = (\alpha_1 \alpha_2 \alpha_3 \alpha_6 \alpha_7)^{128}.$$

6.3 Generating function

For indeterminate t ,

$$G(t) := \sum_{b \in \{0,1\}^8} t^{\log R(b)} = \prod_{i=0}^7 (1 + t^{\log \alpha_i}),$$

which encodes the multiset of $\log R(b)$ additively.

7) Canonical representatives

A canonical choice for each class is obtained by fixing

$$\tilde{b}_0 = 0, \quad e \in \{-1, 0, 1\} \text{ chosen explicitly, } \tilde{b}_i \in \{0, 1\} \ (i \in \{1, 2, 3, 6, 7\}).$$

One may realize e by $(\tilde{b}_4, \tilde{b}_5) = (0, 0), (1, 0), (0, 1)$ for $e = 0, -1, +1$ respectively. Then

$$R(\tilde{b}) = \alpha_1^{\tilde{b}_1} \alpha_2^{\tilde{b}_2} \alpha_3^{\tilde{b}_3} \alpha_6^{\tilde{b}_6} \alpha_7^{\tilde{b}_7} \alpha_4^{-e}.$$

This yields exactly 96 canonical values.

8) Optional alignment functionals

Given a target list $\{y_k\} \subset \mathbb{R}_{>0}$, define alignment scores

$$\Delta_k(b) := |f(R(b)) - y_k|,$$

for a chosen monotone map f (e.g., $f(x) = x$ or $f(x) = c \log x$ with unit constant c). Minimizers of Δ_k are well-defined on the canonical representatives above.

9) Notes on computation

- All aggregate identities hold symbolically; no floating-point evaluation is required.
- When numerics are desired, evaluate $T, \varphi, \pi, \alpha_6, \alpha_7$ at the chosen precision and then form the 96 canonical values; the remaining 160 values are generated by the two symmetries.

Summary: The model is fully characterized by eight positive parameters with two exact relations ($\alpha_0 = 1$, $\alpha_4 \alpha_5 = 1$). The resonance image has size 96, with a 64/32 split between classes of degeneracy 2 and 4. Exact closed forms are provided for global sums and products, and a canonical parametrization enables unambiguous enumeration and alignment.